

Schrödinger's Beat: An XR Rhythm Game for Learning Quantum Computing

Brady Phelps*
Ohio University

Chang Liu†
Ohio University

Chad Mourning‡
Ohio University

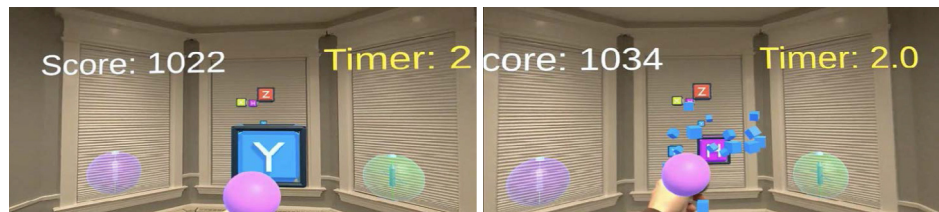


Figure 1: Images seen from the POV of a user in Schrödinger's Beat. The left picture features the user about to smash the qubit, and the after on the right. The qubit smashes into many tiny cube fragments to add to the satisfaction of the game. Additionally, in both images, one can observe the user qubit (purple) and the target qubit (right), as well as the score and timer overhead.

ABSTRACT

Schrödinger's Beat aims to make quantum education intuitive, fun, and active. This research iterates on existing rhythm game concepts and introduces quantum computing principles. Users will follow a brief tutorial, before engaging in the main application, where they use their hands to smash a series quantum gates in time with the rhythm, and purposefully dodge gates to apply them to a qubit. The users aim to smash gates while dodging correct gates to match their qubit state with a goal state. Users gain familiarity with quantum concepts while staying active and interacting with intuitive 3D representations of quantum computing concepts.

Keywords: Extended Reality, Education, Quantum Computing, Games.

1 PROBLEM AND MOTIVATION

Quantum computing promises to revolutionize various fields, including cryptography [12], medicine [19], and optimization [10]. The field was recognized by presidential directives [17] and listed in the NSF's "10 Big Ideas" [11]. Despite its importance, the complex behavior of quantum information presents significant challenges to widespread understanding and adoption. The United States, while a leader in quantum research, faces a shortage of upcoming talent, which poses a national security threat [2]. As such, this introduces a need for initiatives which effectively introduce and engage future scientists and engineers to the world of quantum computing.

2 BACKGROUND AND RELATED WORK

Steep learning curves are not unique to quantum computing, although relating quantum physics to physical reality presents a new challenge [9]. Previous instances of educational software tools have demonstrated utility in abstracted learning systems that simplify information for beginning students. Examples of this include programs like Scratch [13] and Blockly [6], which have taught millions of students programming concepts through simplified programs.

*e-mail: bp309420@ohio.edu

†e-mail: liuc@ohio.edu

‡e-mail: mourning@ohio.edu

Visualizations can dramatically improve the understanding of science education [20], but standard techniques can struggle with the limitations provided by a computer screen. Researchers have recognized the importance of XR and VR visualizations in helping people understand quantum computing concepts; researchers recently created a virtual reality interface to explain historic quantum-mechanical research experiments [16]. Through their research, they successfully modeled quantum mechanics to physical realities. Furthermore, in August 2023, the NSF funded a multi-university project to pursue advancements in quantum education via VR technologies [18]. Their project creates VR environments to educate and share quantum information concepts with a beginner audience.

Games have also been proven to be an effective tool for education [8]. There are many existing games made for the purpose of quantum education, such as Quantum Flytrap [4] or Quantum Odyssey [5]. There has also been instances of VR games geared to teach users about quantum computing, such as QuantumVR [15] and Stranded [14]. The power of games in conveying quantum knowledge has also been recognized by a number of events, such as the Quantum Game Jam [7].

Active participation is also an important and often overlooked factor in education. Physical activity is proven to improve cognition, metacognition [21], and educational performance [1]. The project presented in this paper combines aspects of exercise, games, 3D representations, and XR to create a novel and engaging educational application to introduce beginners to the world of quantum computing.

3 METHODOLOGY

Schrödinger's Beat bridges existing educational gaps by leveraging XR principles to create an active, engaging, and fun medium for learning and interacting with quantum computing concepts. XR as a medium for quantum education has limited prior research, and there has yet to be any published XR or VR rhythm games for quantum educational purposes.

3.1 Game Introduction

To familiarize players with the game environment, a brief introduction is provided. During this introduction, players can physically interact with quantum gates. Players use standard hand controller devices to grab and observe gates. Players will also be able to observe their own "qubit", a 3D Bloch sphere representation. Players are able to grab and view the qubit from all sides and angles, providing them with a level of detail and engagement that cannot be found

in typical 2D or 3D qubit representations. To conclude the tutorial, players enter a slowed tutorial round so they can acclimate to the game.

Once in the game environment, users will see several digital objects in their real-world pass-through environment. They will see their own purple qubit; this qubit is the one they control. They will then see the green target qubit on the right; this qubit holds the state that they are trying to match. Both of these qubits can be observed in Figure 1. A timer is included in the top right, and a score counter in the top left. The timer counts down from 10 seconds, and once the timer ends, the game checks the user's qubit and the target qubit to determine if they match. If they match, the user will get a bonus, as well as a streak multiplier, for the amount of correct matches they get in a row. The user will also be able to see their own body, with pink and blue digital spheres attached to their hands to indicate eligible points of contact within the digital environment.

The music starts shortly after entering the game. The main game currently includes a single custom track that plays at 90 BPM. When the audio starts, a series of quantum gates are spawned on beat across a random array of positions in front of the user. These gates travel towards the user, and the user must smash these gates by touching them with their controllers to earn points. Upon smashing a gate, it will crumble into pieces, adding to the satisfaction of "smashing" a gate, as seen in Figure 1.

To manipulate their own qubit, the user can strategically dodge gates to apply them. When dodging a gate, a particle effect matching the color of the corresponding gate will appear on their qubit. This gameplay system encourages a strategy of hitting as many gates as possible while only dodging essential ones to align the user qubit with the target qubit and maximize points. This influences users to think critically about the effects of gates on qubits and gain an intuition on quantum behaviors.

Repeated and fast-paced engagement is used to help users develop an intuitive understanding of how each gate affects a qubit. Repetition, a proven method for enhancing knowledge retention [3], is paired with play, music, and rapid decision-making to design an effective learning tool.

4 RESULTS AND CONTRIBUTIONS

The development of Schrödinger's Beat marks advancements in making quantum computing education more understandable, active, and fun. By translating complex quantum principles into a fast-paced game format, this work promotes the development of quantum intuition in players. The game is designed so that players must strategically dodge or smash gates to apply quantum operations; the countdown timer adds an element of urgency, challenging users to think quickly while reinforcing critical concepts in quantum mechanics.

The project expands on existing XR research in training and education, providing a novel rhythm based approach to teaching quantum computing concepts. In employing a fast-paced learning environment, we design the application to build intuition around quantum computing concepts. This research explores the potential benefits of XR, games, and physical activity in instructional technologies, while responding to the United States' growing quantum talent shortage.

5 FUTURE WORK

In the future, the authors aim to test the effectiveness of the application by performing trials on university students. These students are the target demographic of this application, as they will fuel the future quantum work force. Additionally, we hope to gain insights into the impact of games, XR, and physical activity in communicating complicated information through these trials by benchmarking the application against other available alternatives. Additional songs and difficulty levels may also be added.

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